

API Automation Framework Project Report

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Project title

REST Assured API Automation Framework using Java, JUnit 5, and Allure

Objective

The objective of this project is to automate REST API testing using Rest Assured with Java and JUnit 5. The framework is designed in a simple corporate-style structure with separate packages for clients, configuration, constants, models, specifications, and tests. Allure reporting is integrated to generate detailed test execution reports.

Technology stack

Programming Language: Java

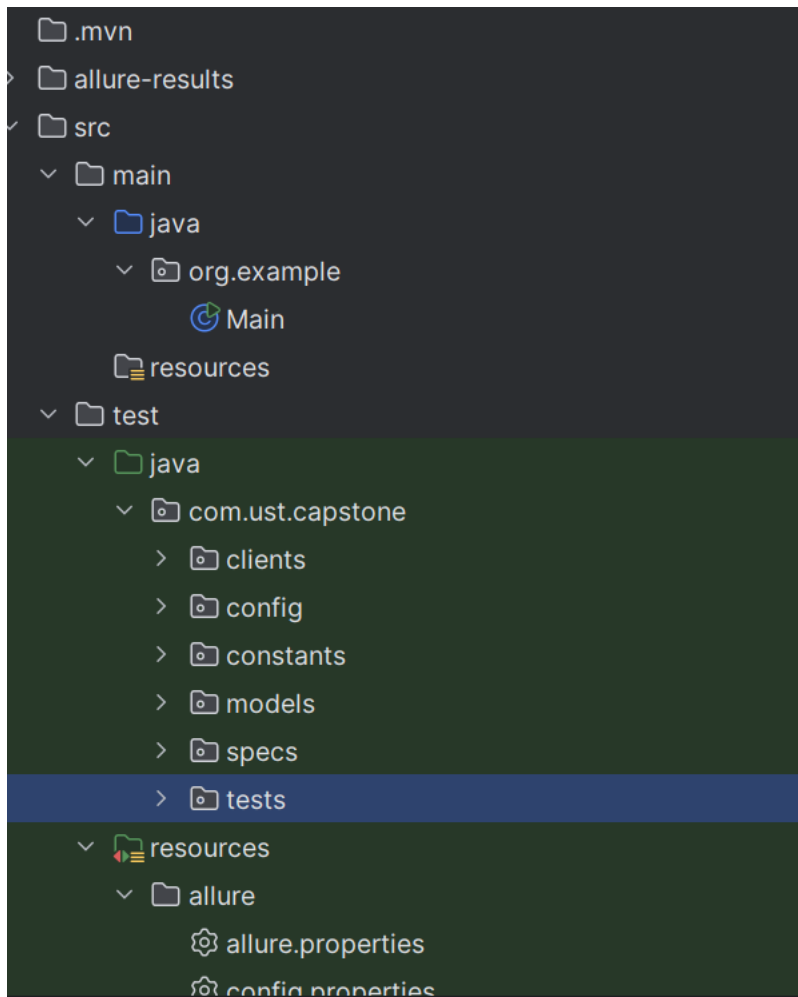
API Testing Tool: Rest Assured

Test Framework: JUnit 5

Build Tool:Maven

Reporting Tool:Allure Report

Framework structure



Package description

Clients

Contains reusable API request methods for Create User, Generate Token, and Get Books APIs.

Config

Contains ConfigReader.java to read values from config.properties.

constants

Stores API endpoint paths in Endpoints.java

models

Contains POJO classes for request and response payloads.

Specs

Contains `RequestSpecs.java` for common RequestSpecification such as base URI and headers.

tests

Contains `DemoQATests.java` with end-to-end API test scenarios.

Resources

Contains configuration files such as `config.properties` and `allure.properties`.

API scenario automated

The following API workflow was automated using the DemoQA Book Store API:

1. Create User – POST /Account/v1/User
2. Generate Token – POST /Account/v1/GenerateToken
3. Get Books – GET /BookStore/v1/Books

Test flow

1. Read the base URL from `config.properties`.
2. Create a new user using Rest Assured.
3. Verify the Create User API returns status code 201.
4. Generate an authentication token for the created user.
5. Verify the Generate Token API returns status code 200.
6. Extract the token from the response.
7. Pass the token in the Authorization header.
8. Call the Get Books API.
9. Verify the Get Books API returns status code 200.
10. Log all request and response details using `.log().all()`.

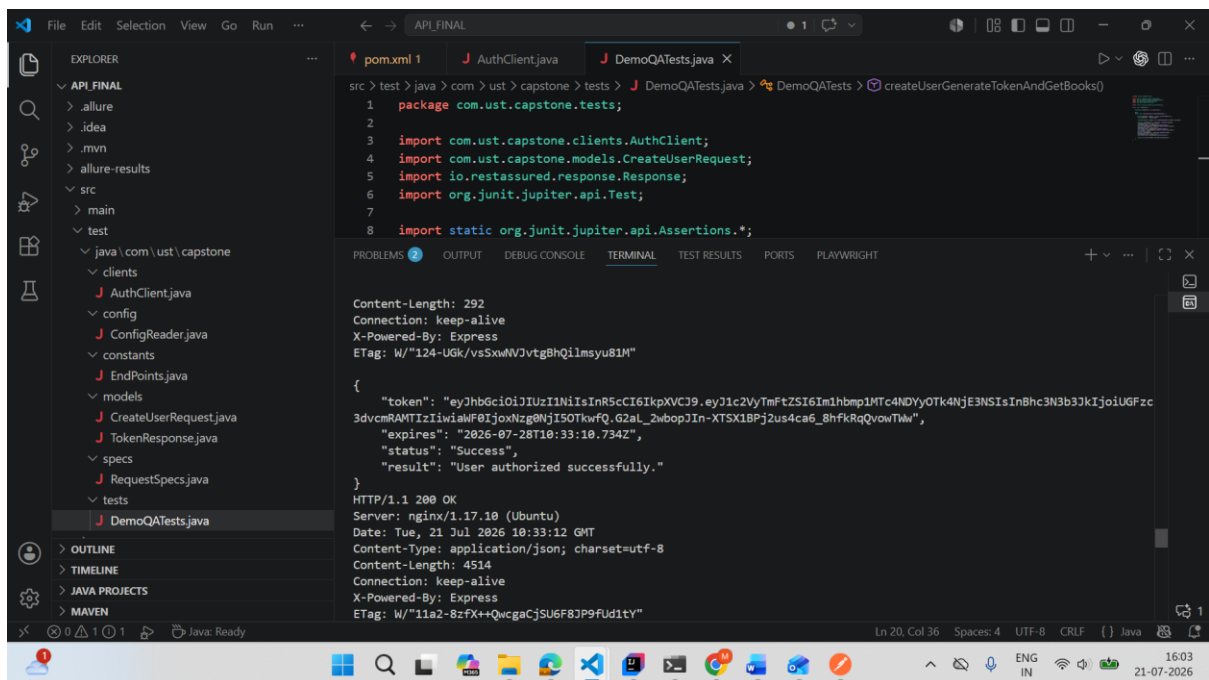
Run tests:

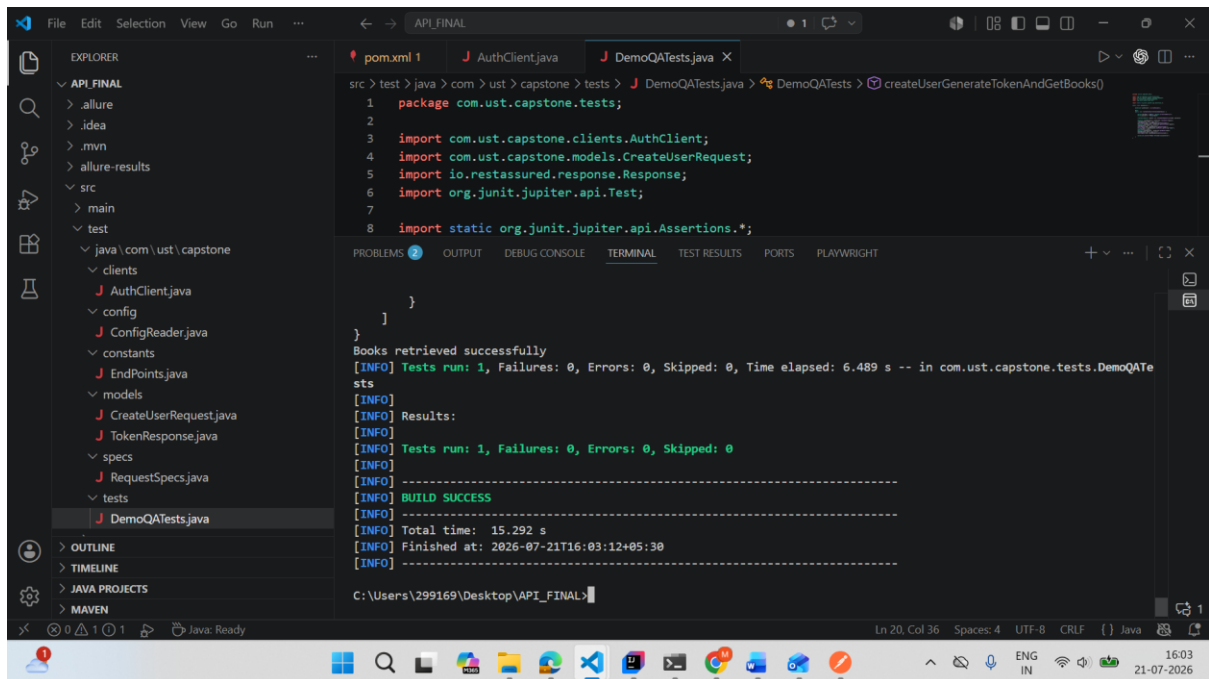
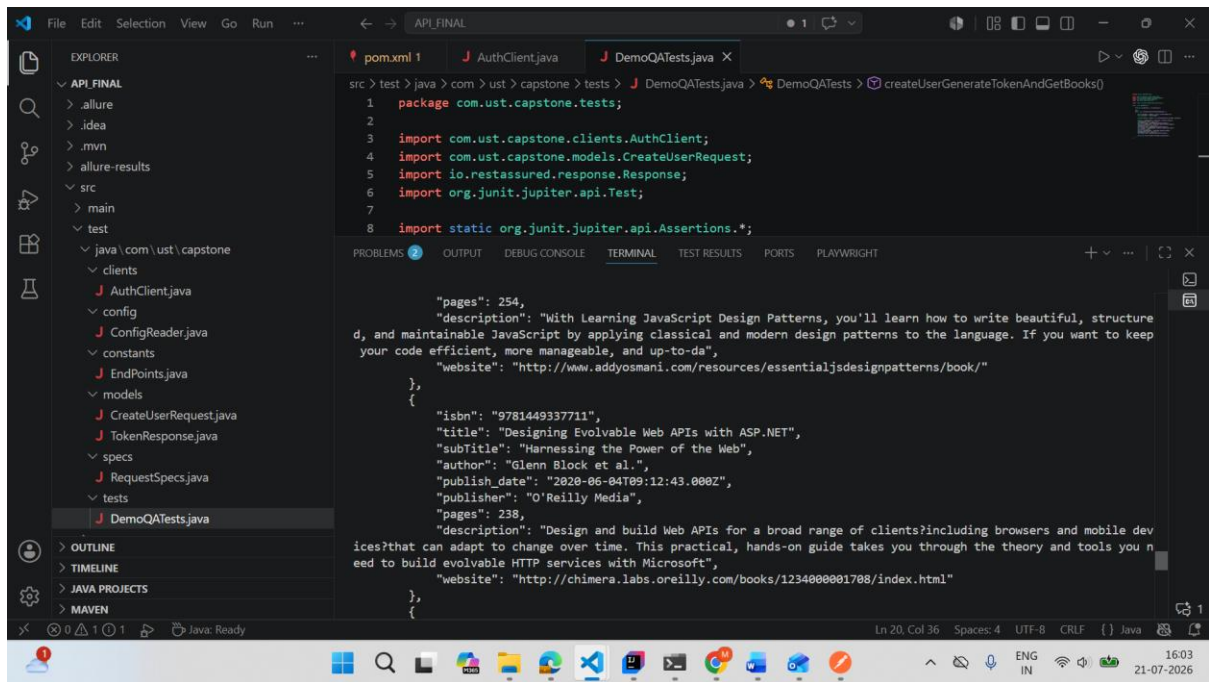
mvn clean test

Conclusion

The REST Assured API Automation Framework successfully automates the Create User, Generate Token, and Get Books API workflow. The framework follows corporate-

Evidences





allure-results

- { 0a6763a8-8865-4e89-9f37-329be0cf0610-result.json
- { 0adc309d-0244-4f45-b401-077a43cd5dcd-container.json
- { 1f4efded-d1d8-4096-afde-e6742dd19333-container.json
- { 2faf9075-b355-47fc-803d-e4d7929f27c0-container.json
- { 2fd1d97f-dc57-417b-9744-e6c0b75321fa-container.json
- { 3b30ca90-8487-4956-8f79-7e2efa10c3db-container.json
- { 3e1d14e1-98ba-4138-8045-bbd93440221b-container.json
- { 3ea25396-9683-4605-b376-67011a443832-result.json
- { 4c35ddb0-5cf3-4b30-b06f-605bf33a362b-container.json
- { 4d83f533-d352-4508-9fb2-59bf6c747efb-result.json
- { 4d85e0b1-7c9c-4ba4-af9e-bd92d4e9f44e-container.json
- { 4d665df1-53d1-4484-99e7-c1a26d05ccca-container.json
- { 4da05279-6a86-4688-94dc-98bbb4f61096-container.json
- { 4eea6566-9c64-4795-a51e-4f44b1bcb7f9-container.json
- { 5c6446fe-0417-4b16-ad7c-4114bae98a78-result.json
- { 5d8cc5b0-00fa-44d6-8881-4e22d2d32ccf-container.json
- { 8eb104f8-f503-45ab-ab55-4ee03dd6d780-container.json
- { 9c1888e7-be48-47ec-9ac5-75c9d15ce795-container.json
- { 9e222cc5-c908-48c3-bfc6-1665ab0cae5a-container.json